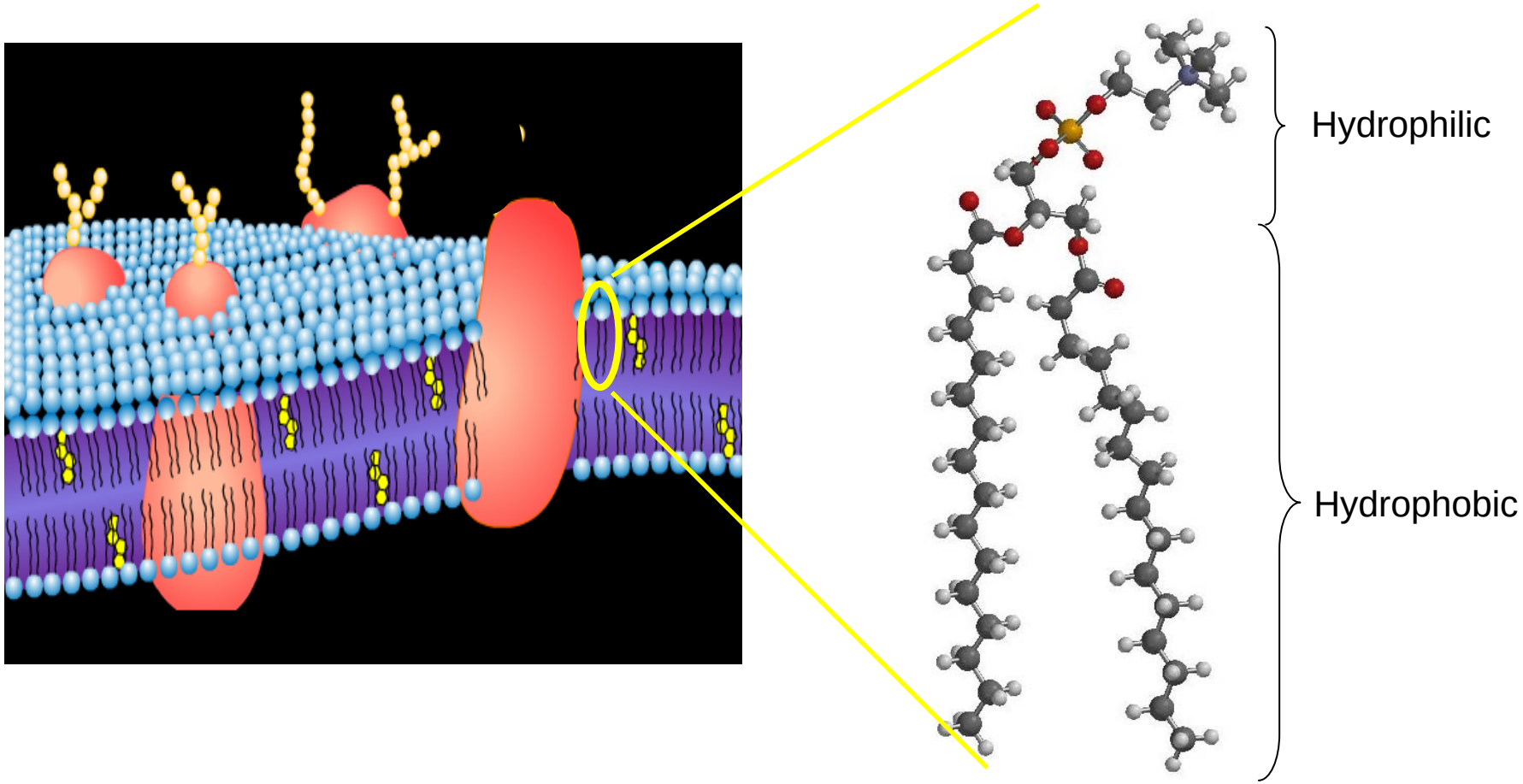
화학특강 (2008년 2학기)
제3장

황현석

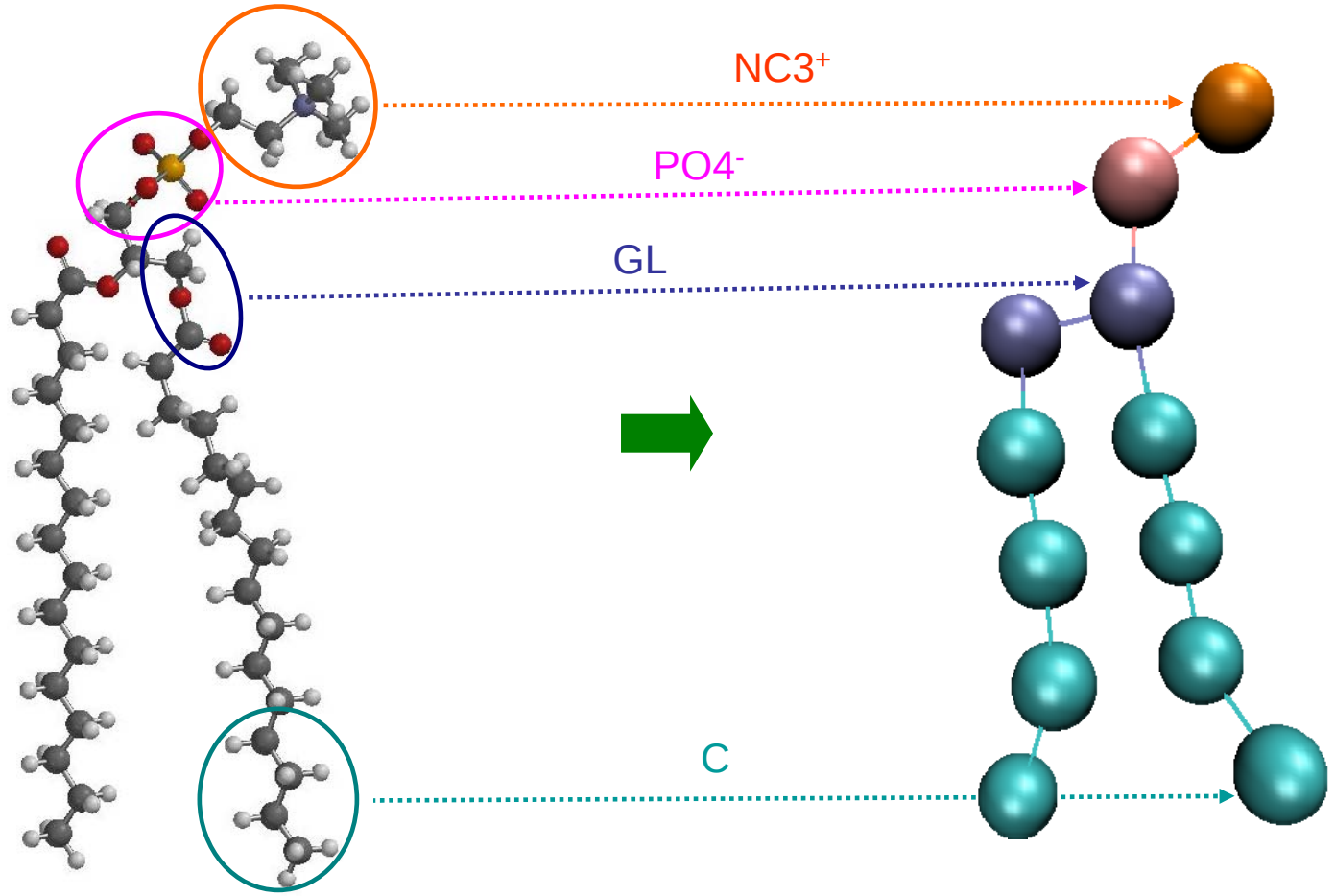
2008년 10월 6일



Self-Assembly



CG modeling for DPPC lipid molecules



all atom model

coarse grained model

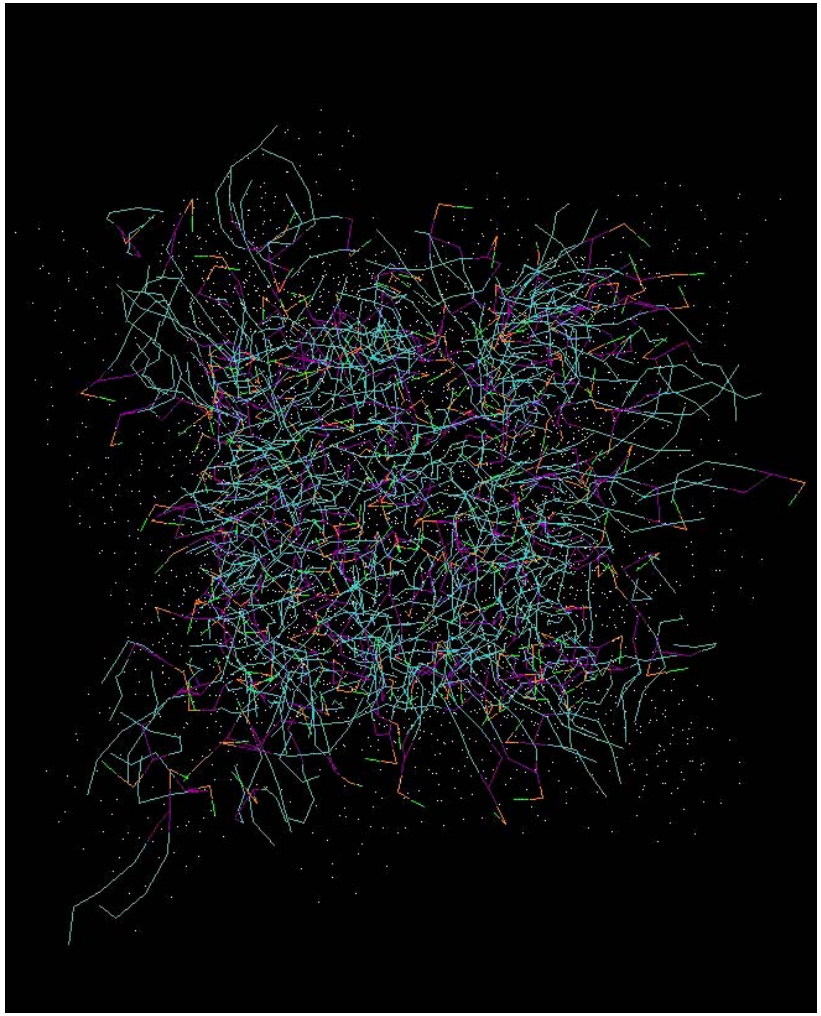
Self-Assembly of Lipid Molecules

- System preparation
- 256 CG DPPC.
- 3072 CG water molecules.
- Isothermal-isobaric ensemble (NPT) :
- pressure : 1 atm.
- temperature : 323 K.
- Periodic boundary condition

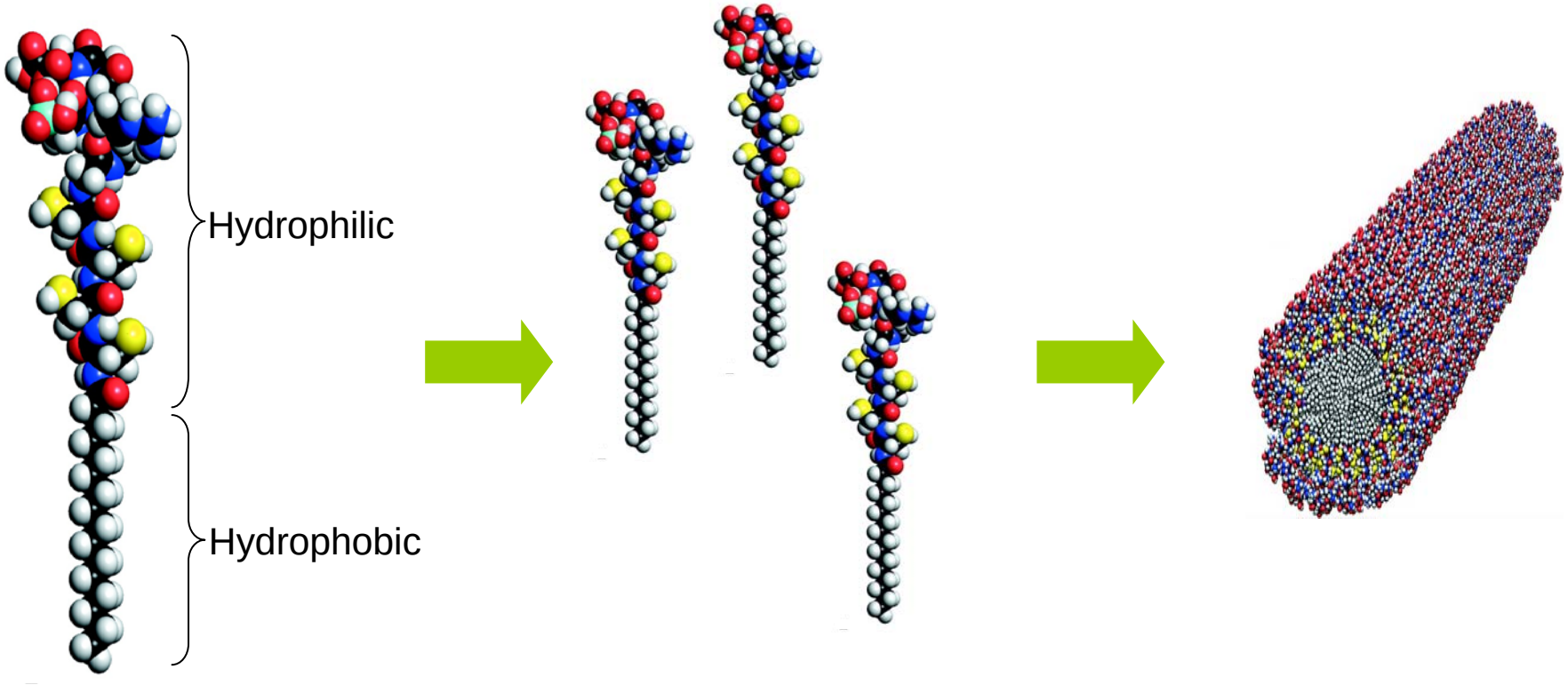
in x,y,z directions.

- Dielectric constant for Coulomb

interaction : $20\varepsilon_0$.

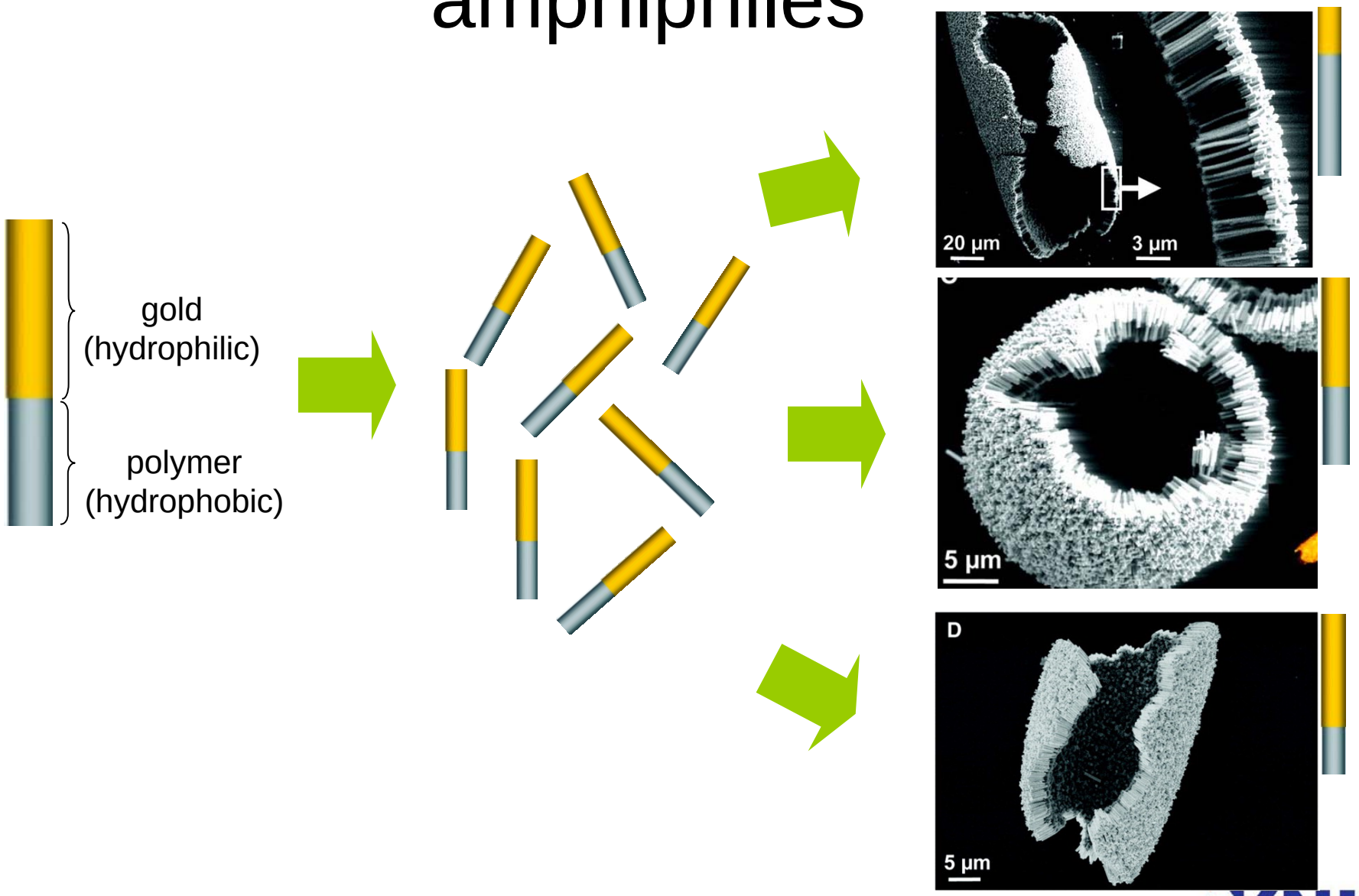


Self-Assembly of Peptide Amphiphile Nanofibers



Hartgerink et al., Science **294**, 1684 (2001)

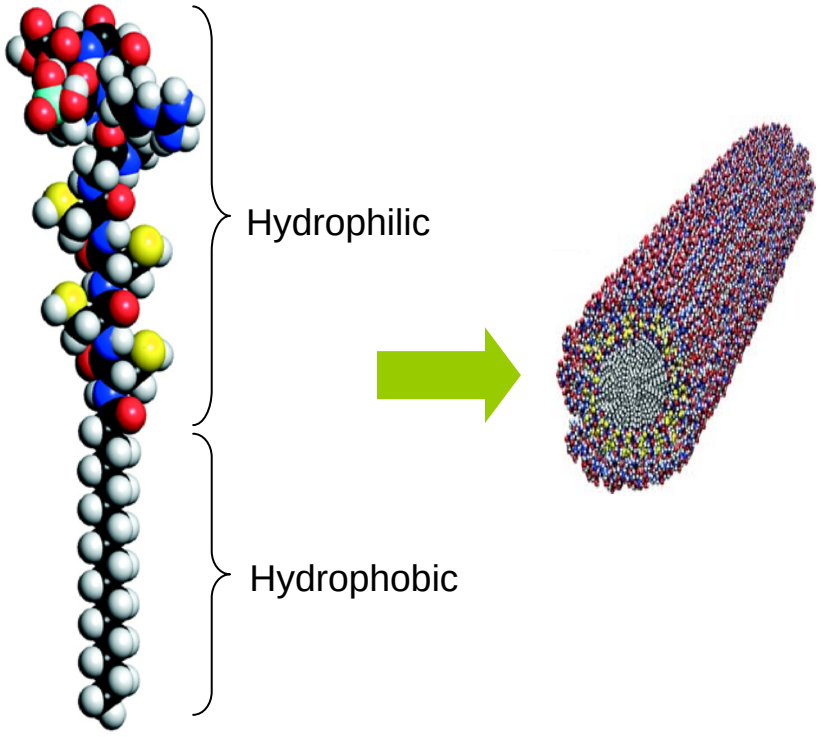
Self-assembly of metal-polymer amphiphiles

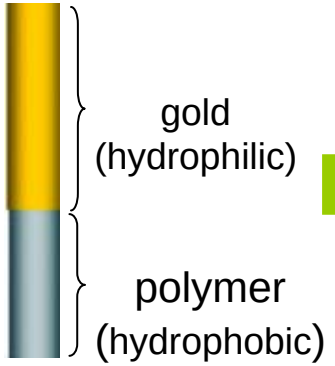


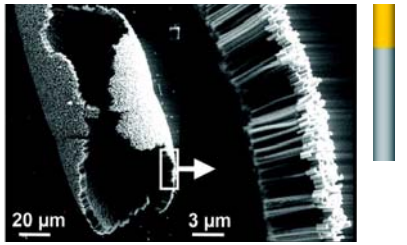
Park et al., Science **303**, 348 (2004)

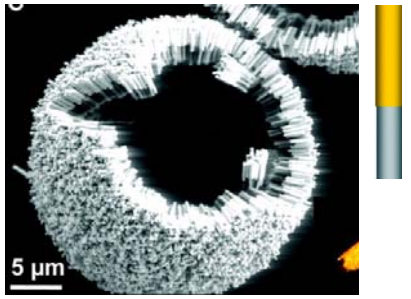
Principles of Self-Assembly

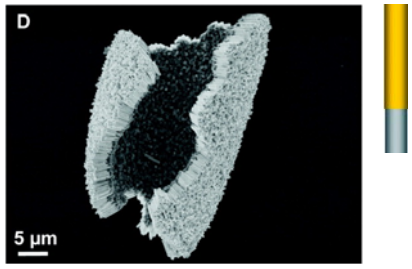
- Components





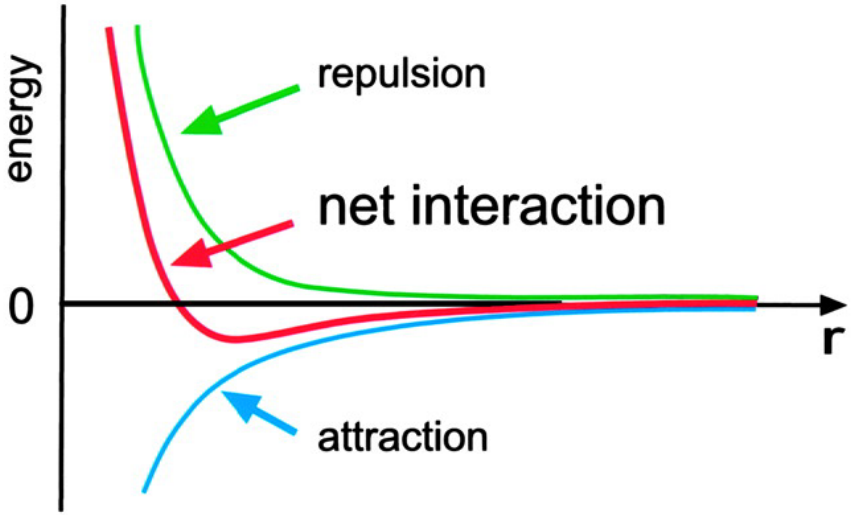


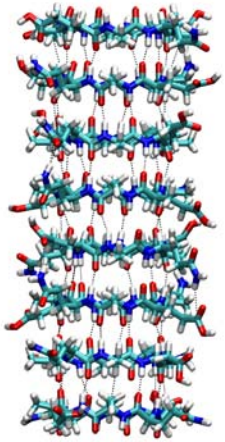


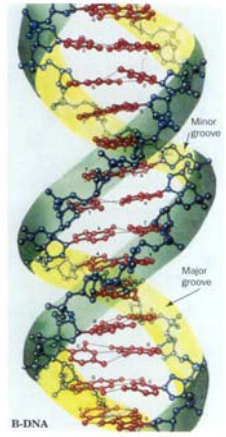


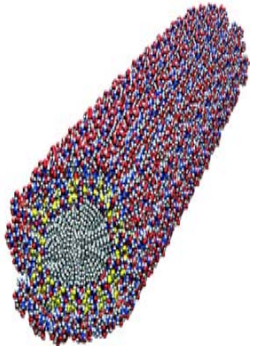
Principles of Self-Assembly

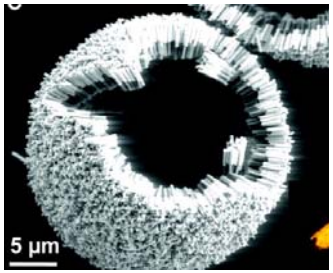
- Interactions







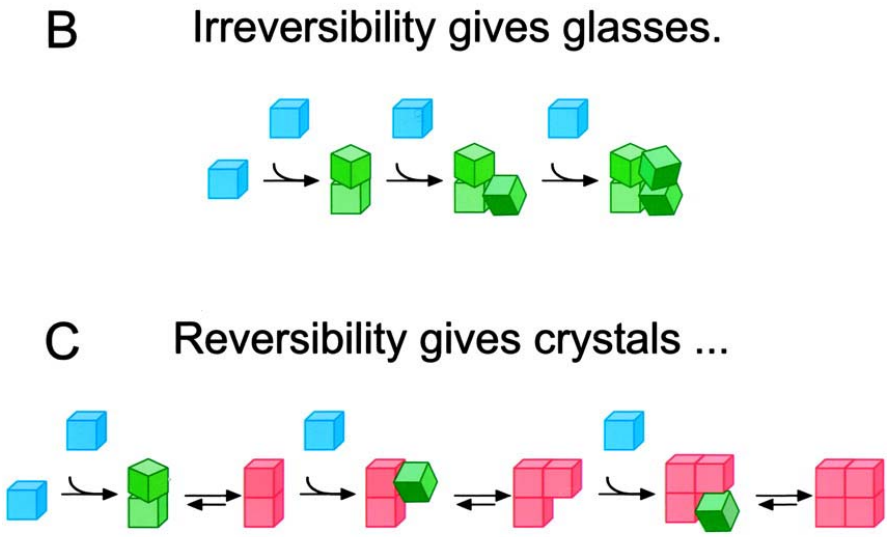




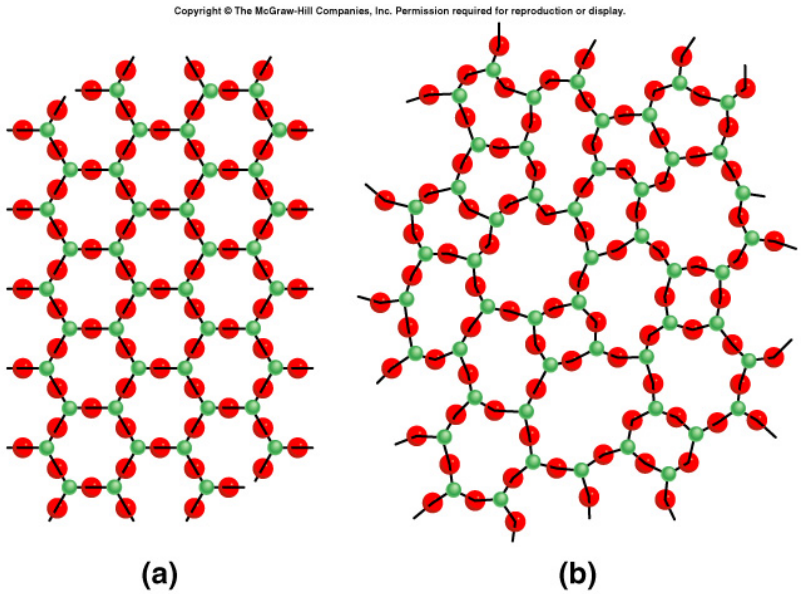
G.M Whitesdies and M. Boncheva, Proc. Nat. Acad. Sci. USA **98**, 4835 (2002)

Principles of Self-Assembly

- Reversibility (equilibrium)



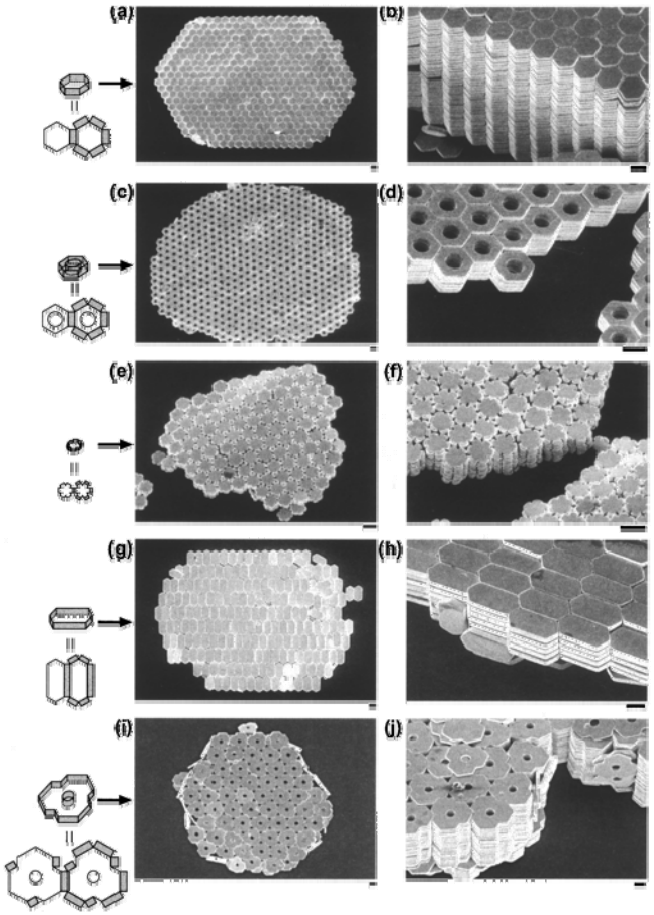
Crystalline quartz and noncrystalline quartz



Chang, 11.7

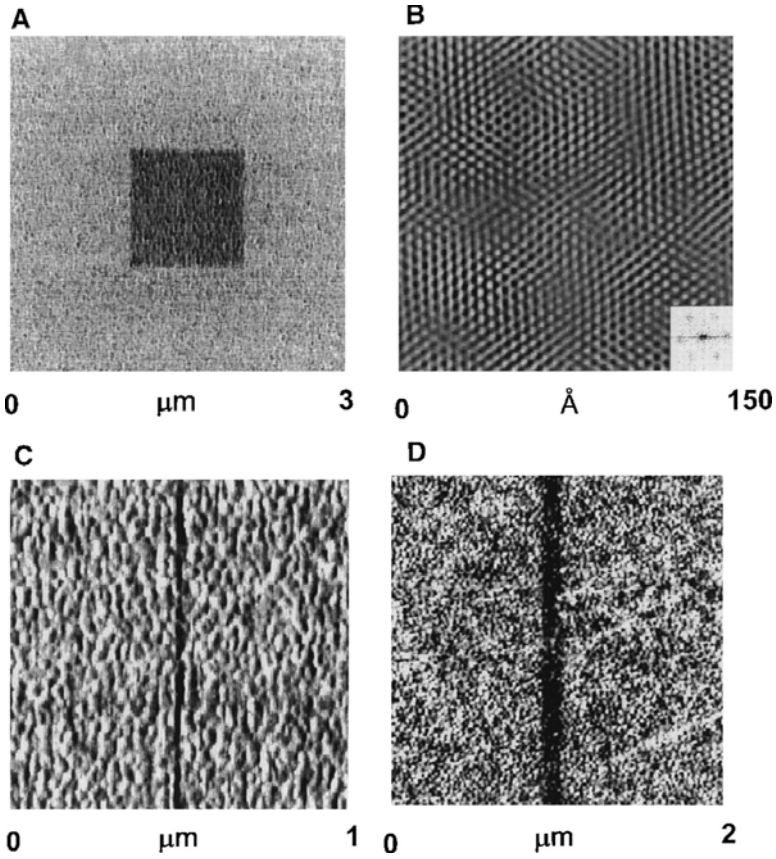
Principles of Self-Assembly

- Environment



Whitesdies et al., JACS **123**, 7677 (2001)

DPN on gold and on mica



Mirkin et al., Science **283**, 661 (1999)

Self-Assembly

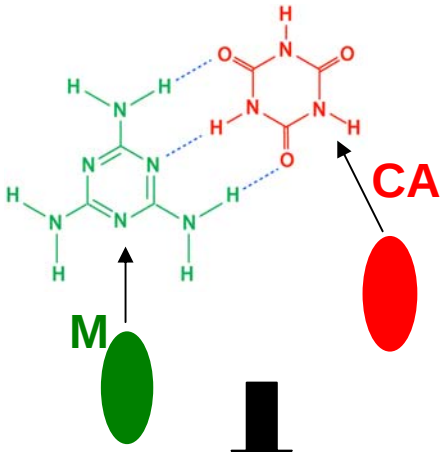
1. Self-assembly is a process in which components, either separate or linked, spontaneously form ordered aggregates.
2. Self-assembly can occur with components having sizes from the molecule (molecular self-assembly) to the macroscopic, provided that appropriate conditions are met.
3. In molecular self-assembly, the molecular structure determines the structure of the assembly.
4. The sucess of self-assembly in a molecular system is determined by five characteristics of the system.
    1) Components: A self-assembling system consists of a group of molecules or segments of macromolecule that interact with one another.

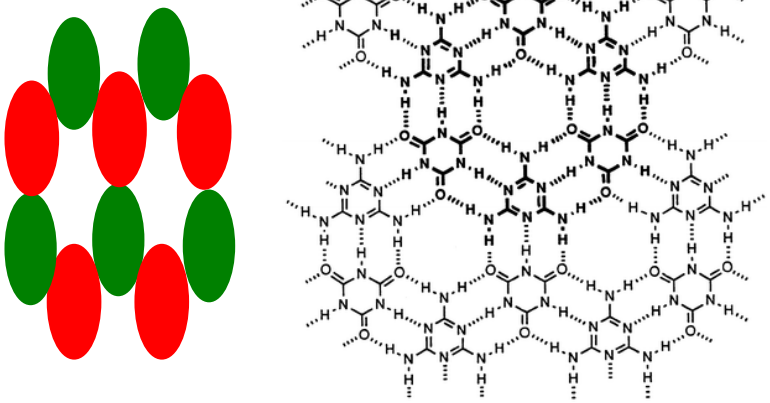
Self-Assembly

2) Interactions: Self-assembly occurs when molecules interact with one another through a balance of attractive and repulsive interactions. This interactions are generally weak and noncovalent (van der Waals, Coulomb interactions, hydrophobic interactions, and hydrogen bonds) interactions.
3) Reversibility (equilibrium): For self-assembly to generate ordered structures, the association either must be reversible or must allow the components to adjust their positions within an aggregate once it has formed.
4) Environment: The self-assembly of molecules normally is carried out in solution or at an interface to allow the required motion of the components.
5) Mass transport and agitation: For self-assembly to occur, the molecules must be mobile. In solution, thermal motion provides the major part of the motion required to bring the molecules into contact.

Molecular Self-Assembly

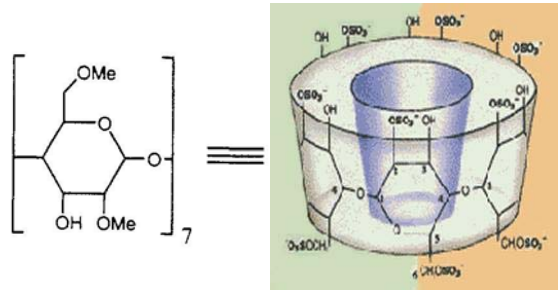
- Melamine (M) - cyanuric acid (CA) complex (hydrogen bonds)

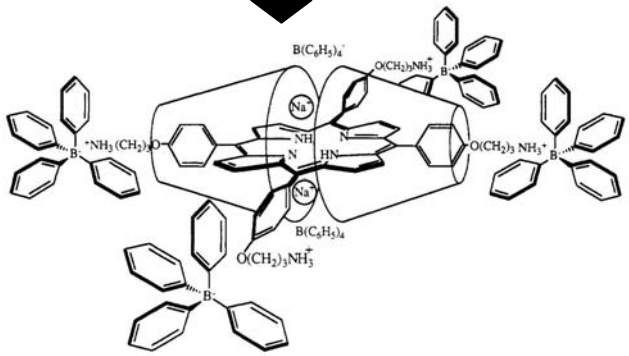




Seto and Whitesdies, JACS **112**, 6409 (1990)

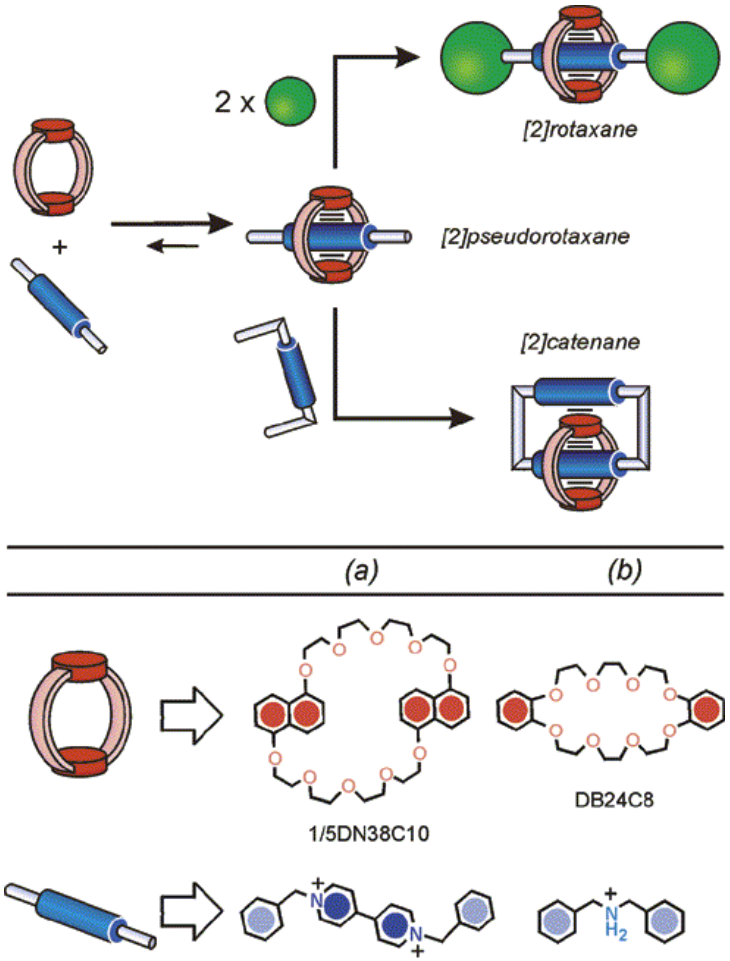
- Supramlecular complex containing porphyrin and β-cyclodextrin (hydrophobic interaction)





Manka and Lawrence, JACS **112**, 2440 (1990)

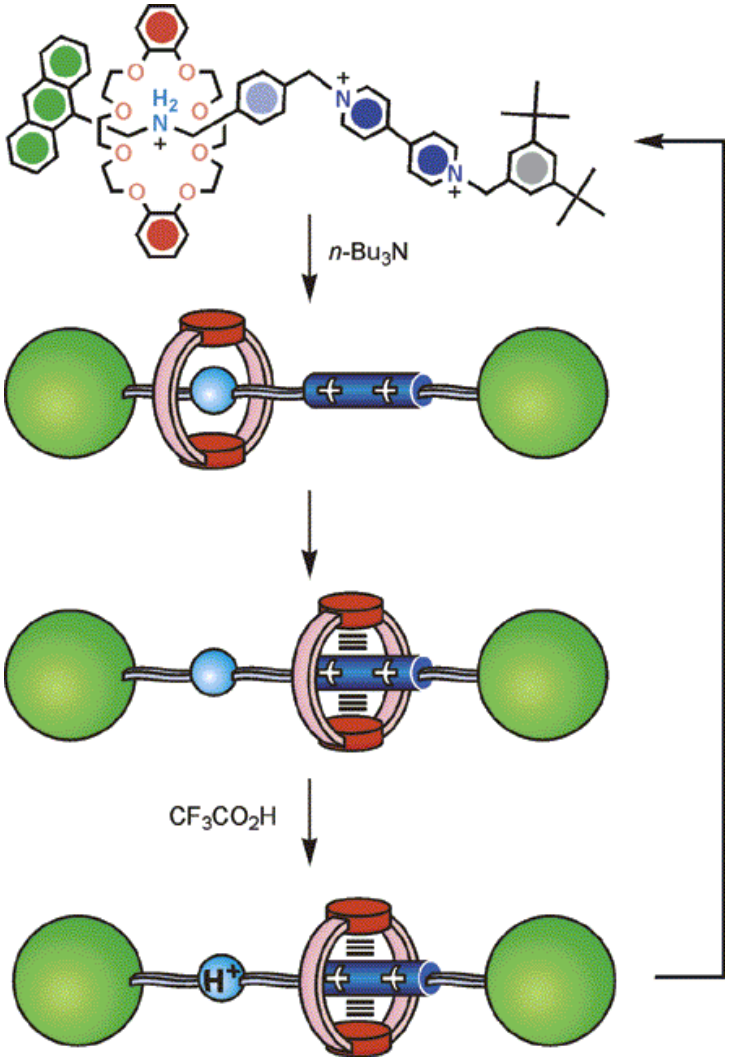
Interlocked Molecules

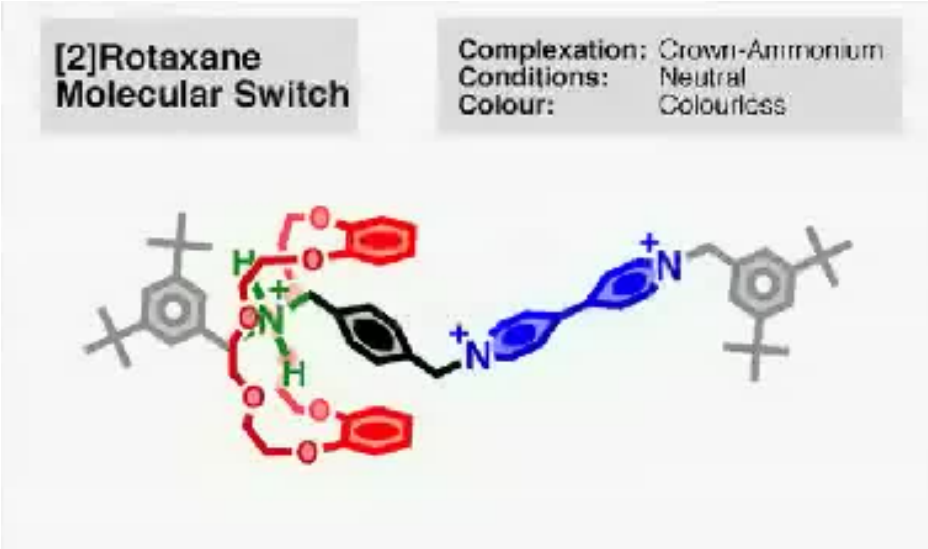


- **Supramolecular chemistry:** the area of chemistry that focuses on the noncovalent bonding interactions of molecules.
- **Rotaxane**: molecular assemblies in which a linear component is mechanically threaded through one or more cyclic molecules with the ends of the linear component terminated with bulky groups.
- **Catenane**: linked supramolecular systems consisting of cyclic subunits held by a mechanical means
- **Host guest adduct**: Interaction between π-electron rich hydroquinone and π-electron deficient pyridinium to form π stack (3.7 Å between rings) + dispersive, electrostatic interactions.

Balzani et al., Acc. Chem. Res. **34**, 445 (2001)

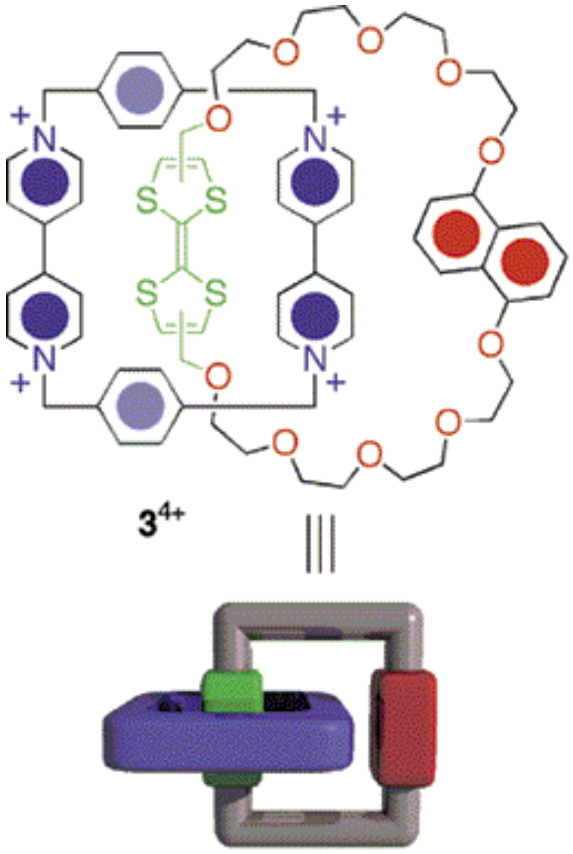
Rotaxanes: A Molecular Shuttle

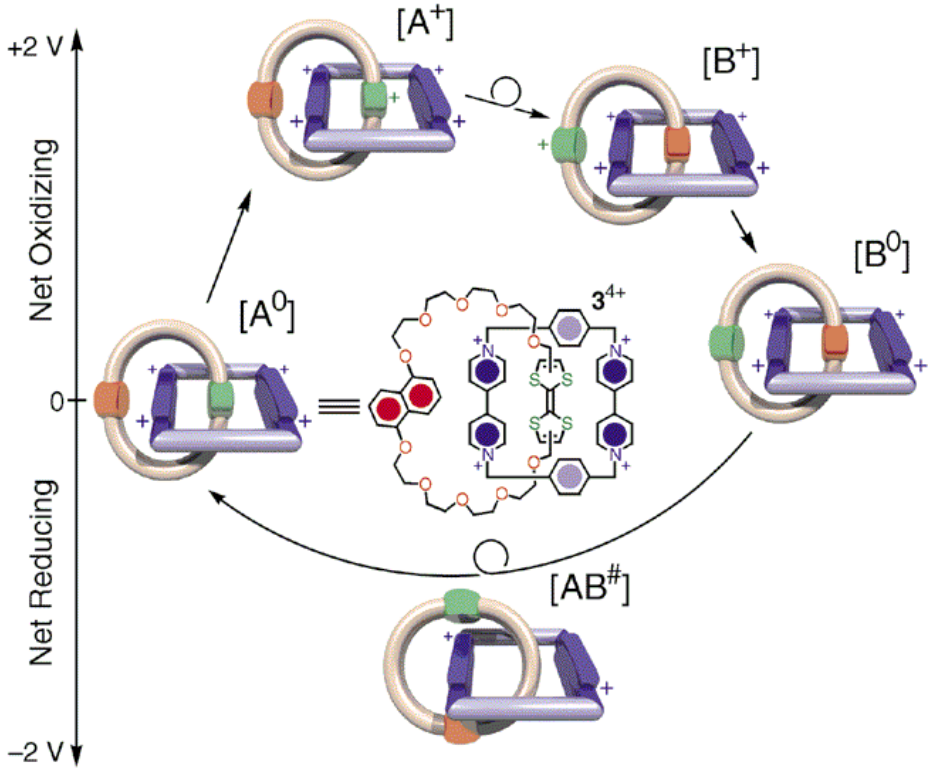




Balzani et al., Acc. Chem. Res. **34**, 445 (2001)

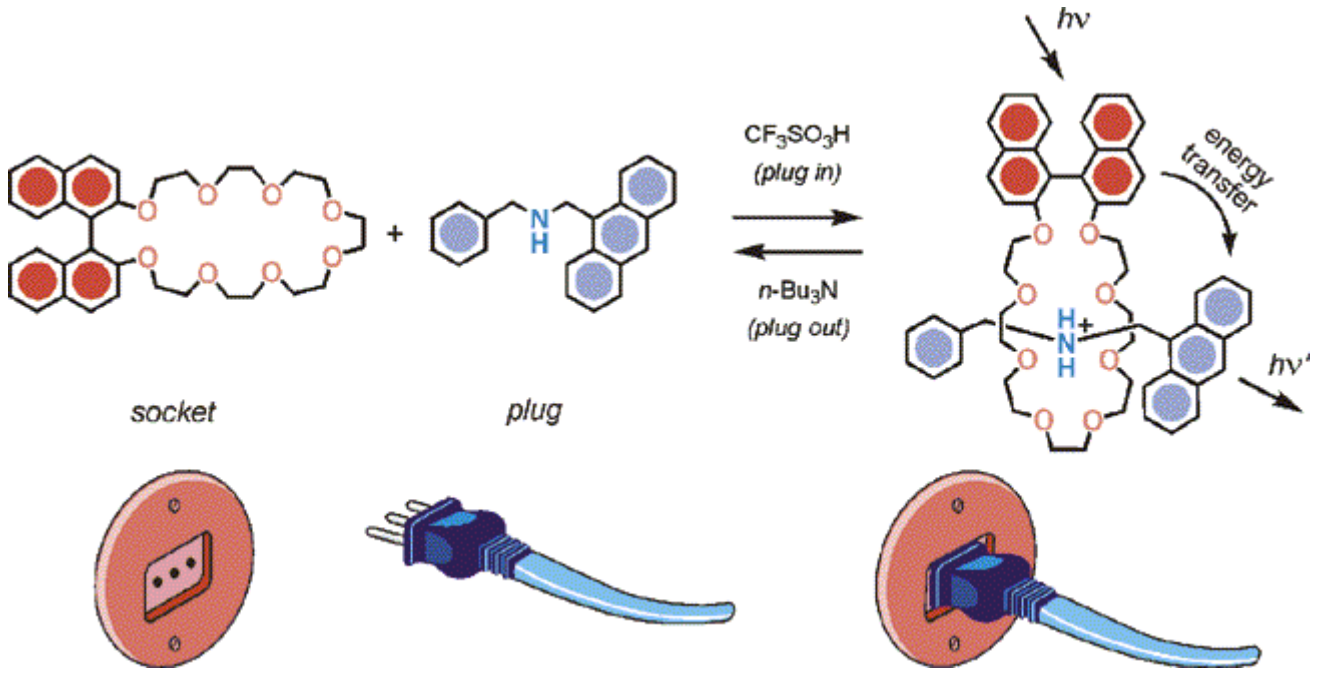
Catenanes: A Molecular Switch





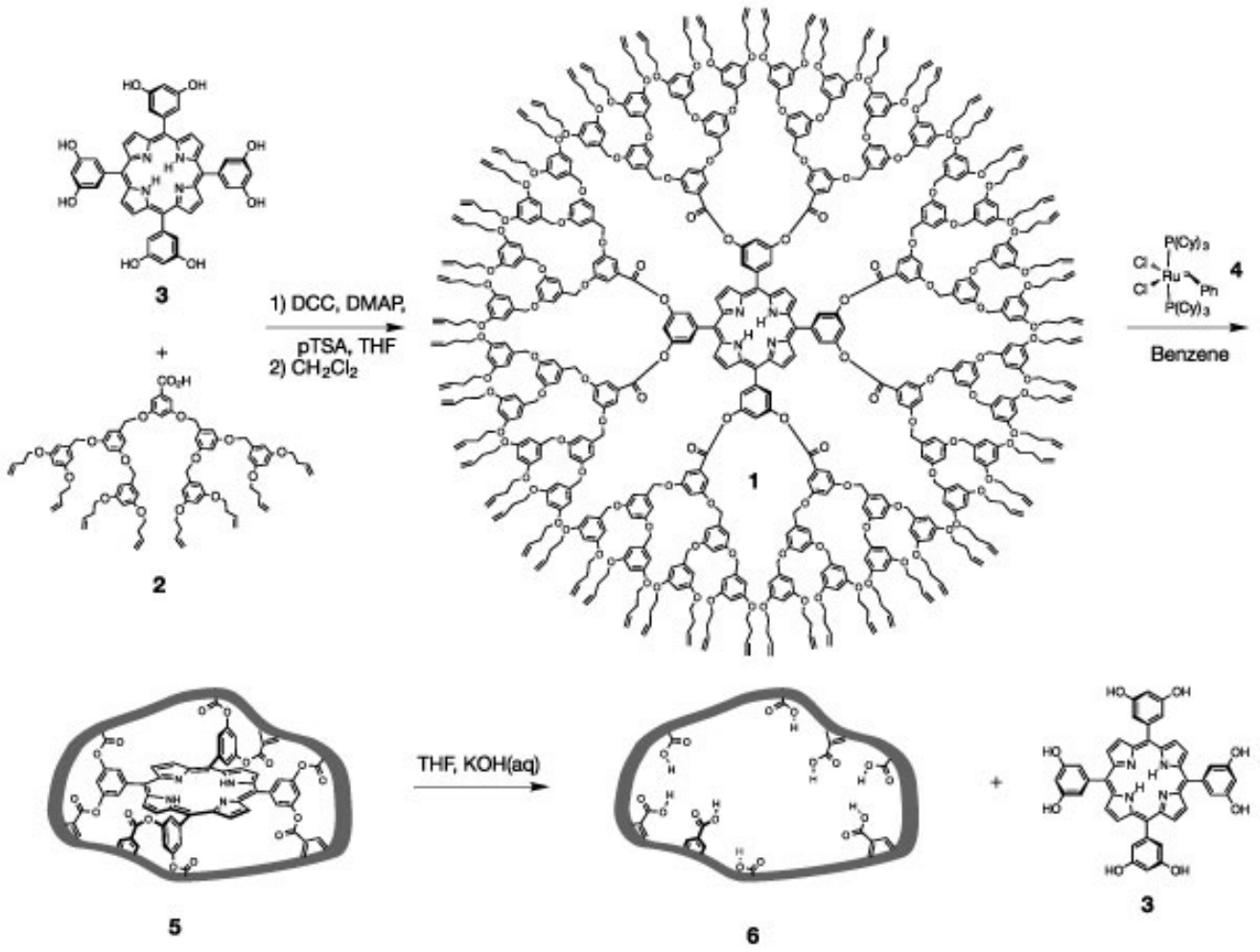
Balzani et al., Acc. Chem. Res. **34**, 445 (2001); Stoddart et al., Acc. Chem. Res. **34**, 433 (2001)

A Molecular Socket-Plug device



Balzani et al., Acc. Chem. Res. **34**, 445 (2001)

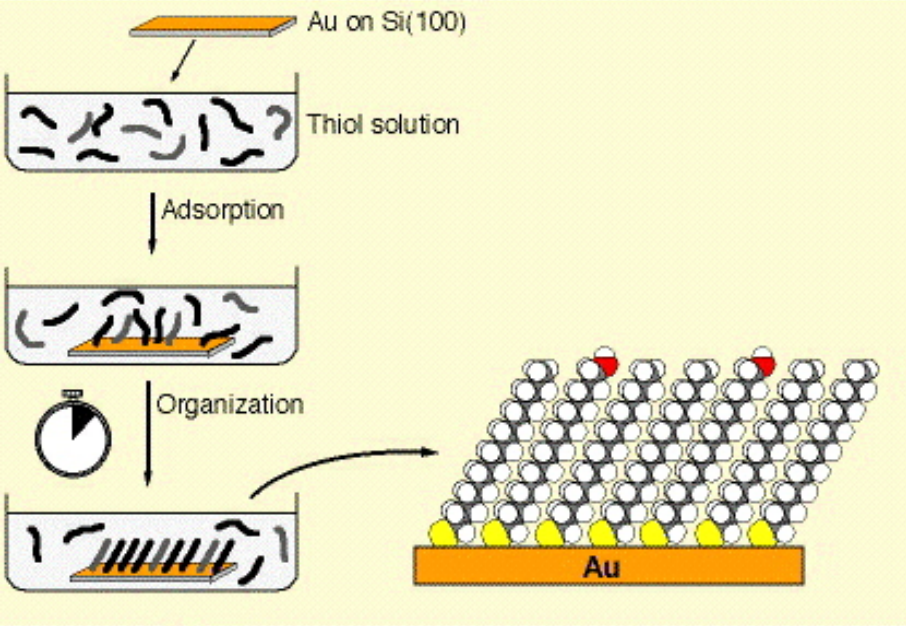
Synthetic Hosts by Molecular Imprinting



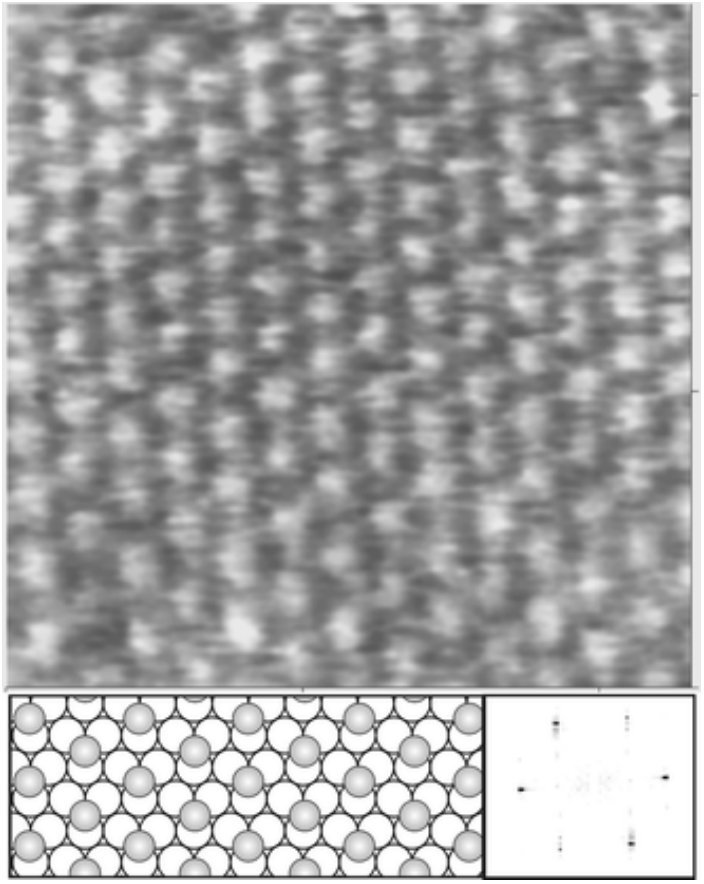
Zimmerna et al., Nature **34**, 399 (2002)

Self-Assembled Monolayers (SAM)

- Alkanethiolates on Au and Ag



Alkanethiolates are tilted approximately 30° from the normal to the surface to maximize the van der Waals interactions between adjacent methylene group

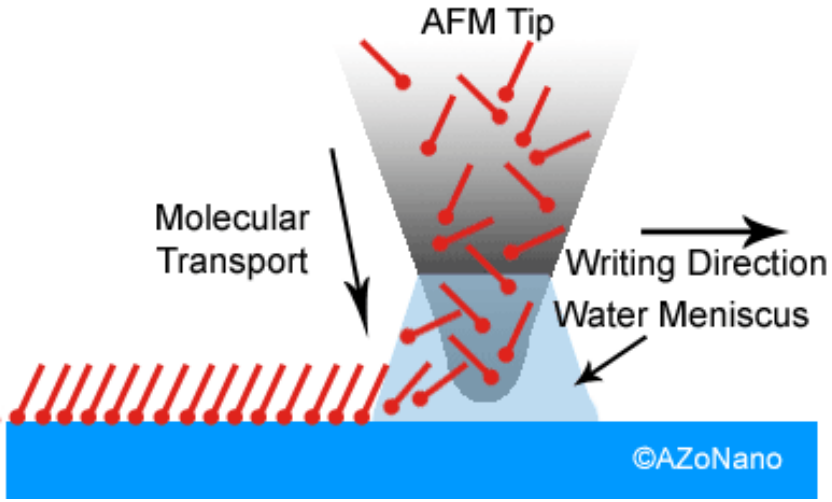


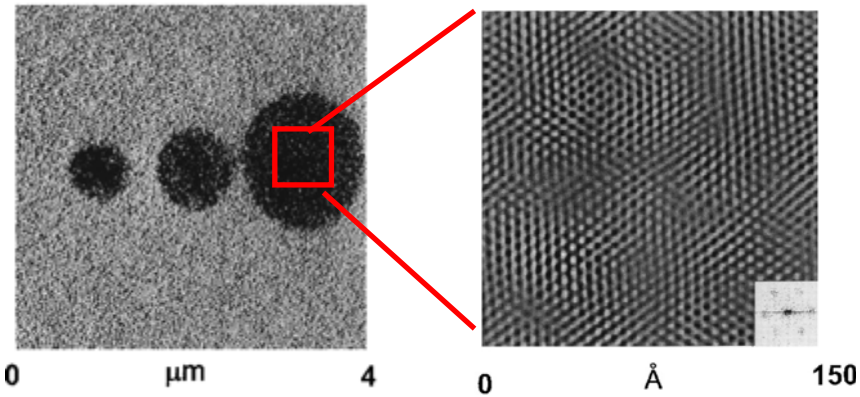
Hexanethiolates on Au[111]

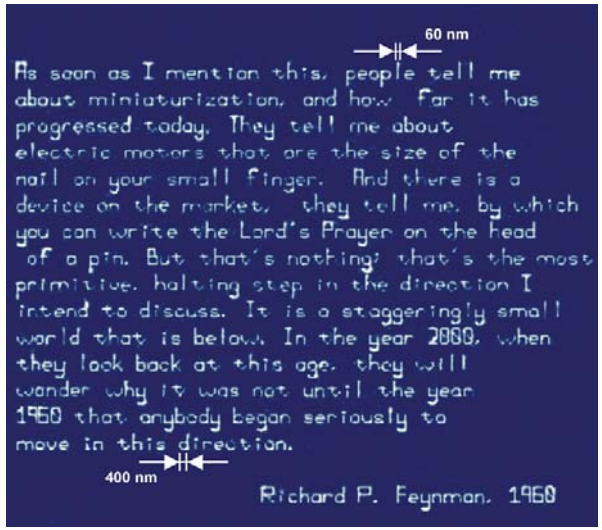
Vericat et al., PCCP **7**, 3258 (2005)

Patterning SAM via Dip Pen Nanolithography (DPN)

- Coated AFM tip is scanned across a substrate, depositing chemical “ink” in its wake.



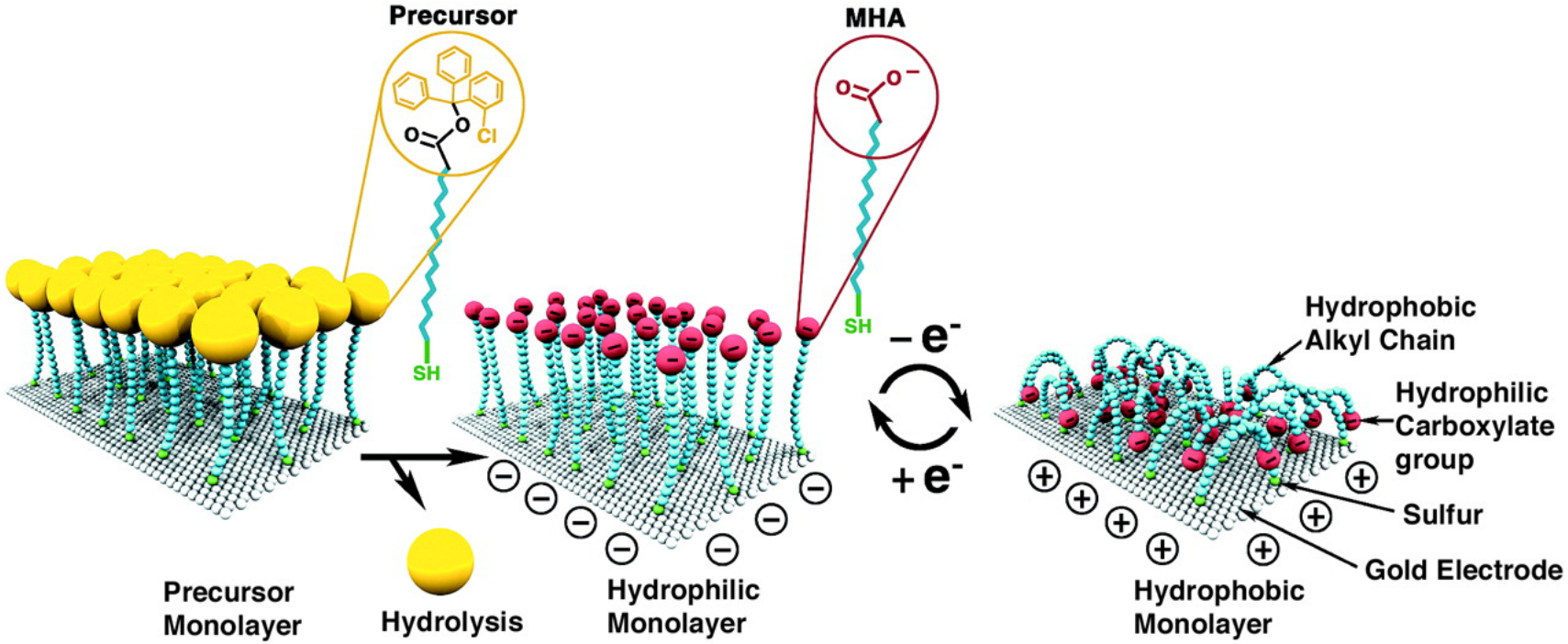




Mirkin et al., Science **283**, 661 (1999)

SAM Applications

- Surface engineering: hydrophilic monolayer ↔ hydrophobic monolayer

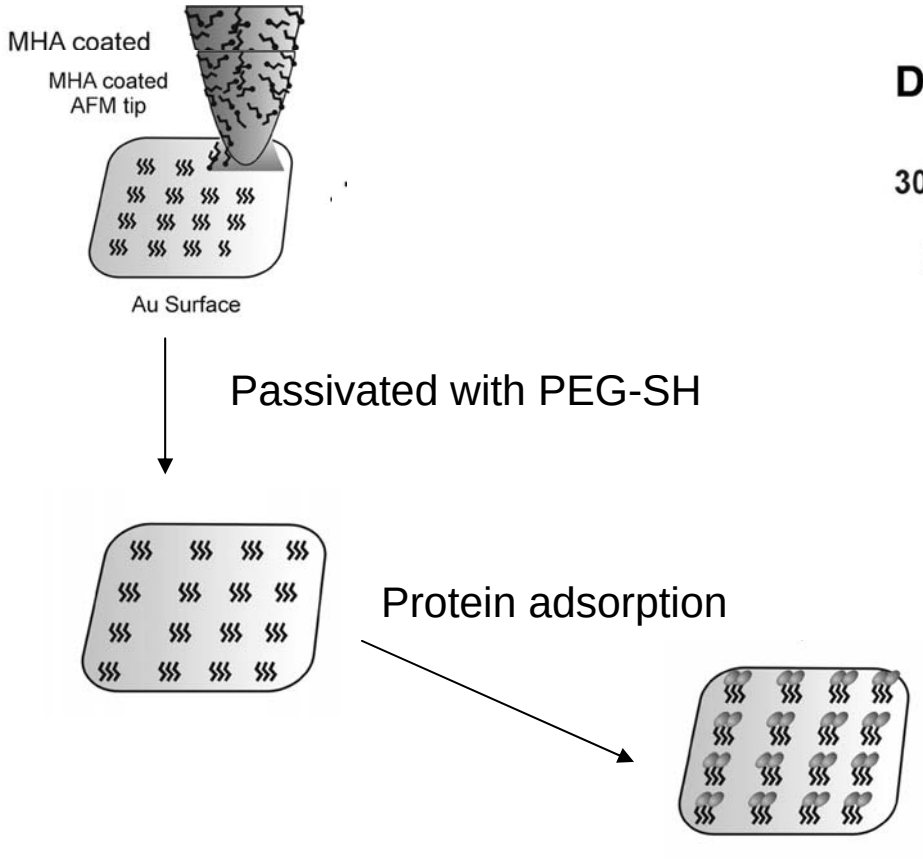


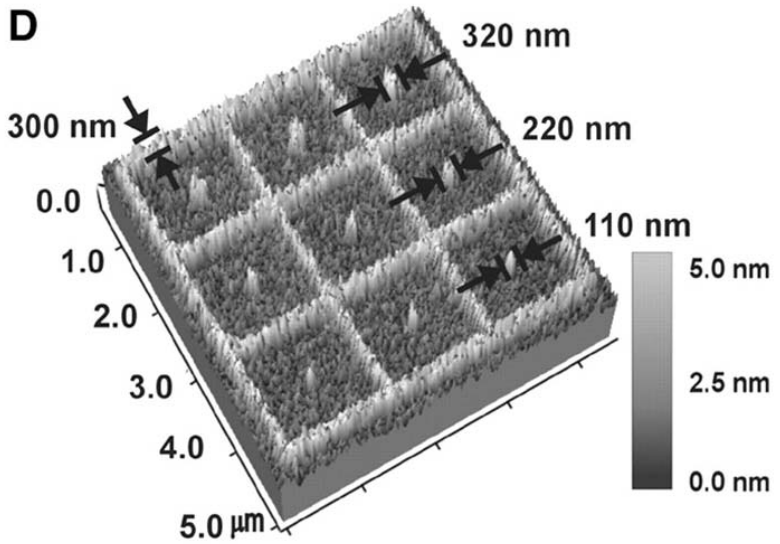
MHA = (16-mercapto)hexadecanoic acid

Lahann et al., Science **299**, 371 (2003)

SAM Applications

- SAM for templates: Biomolecules

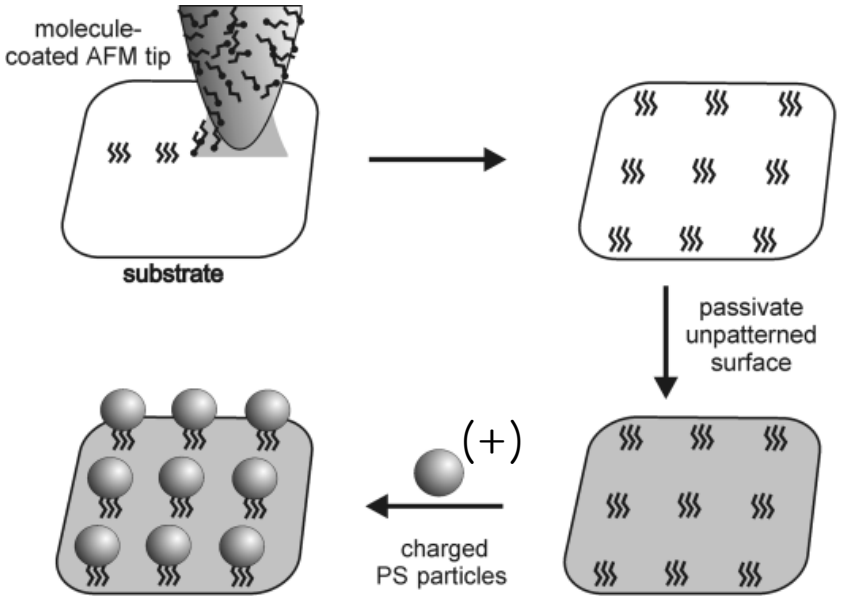


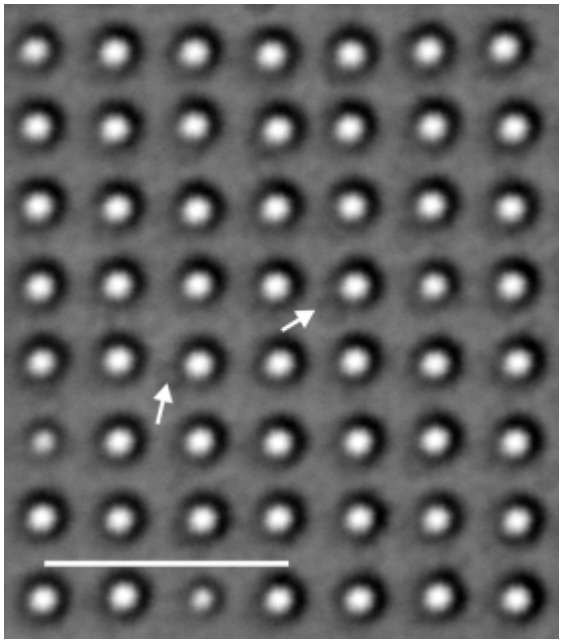


Mirkin et al., Science **295**, 1702 (2002)

SAM Applications

- SAM for templates: electrostatic assembly of charged polystyrene particles





Optical micrograph of 1.0 μm amine-modified PS particles on MHA templates

Mirkin et al., Angew. Chem. Int. Ed. **40**, 3069 (2001)